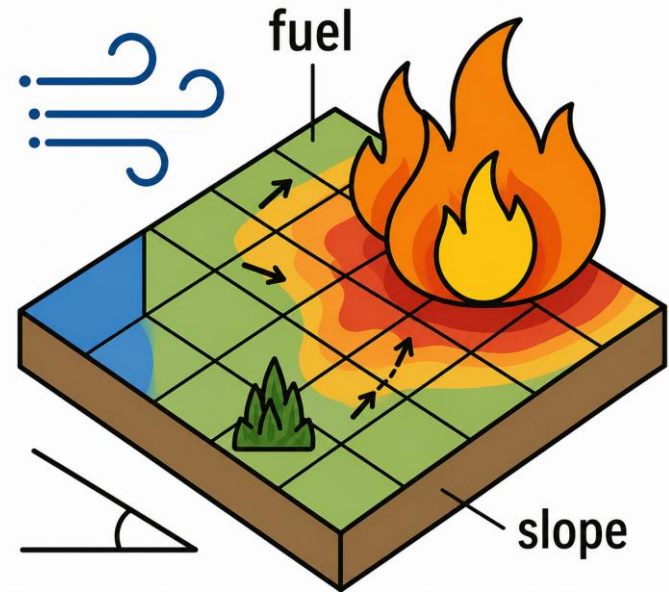


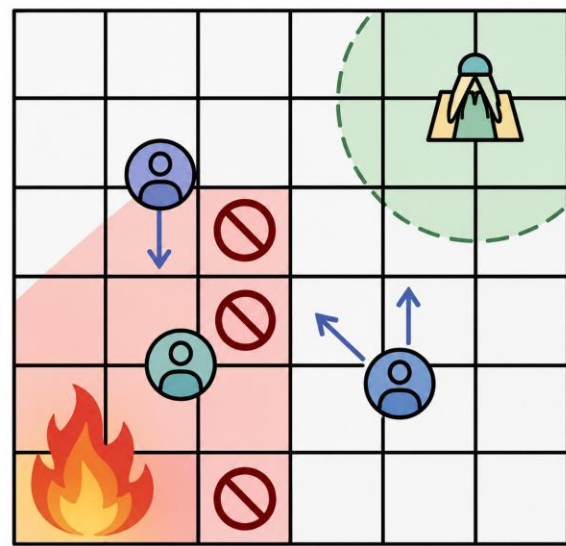
(a) Environment

Cell2Fire engine



Canadian FBP Physics
• Fuel • Wind • Slope
32x32 Grid

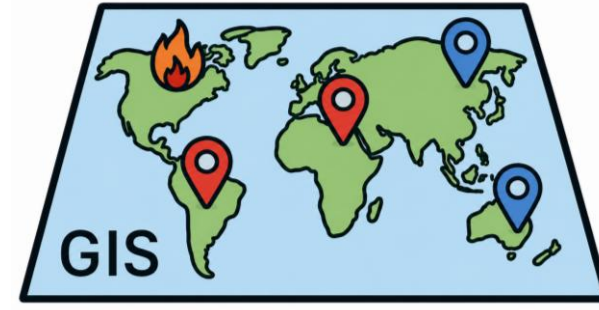
Multi-Agent Wrapper



Suppression Actions
Asset Cascades
Masking (PettingZoo)

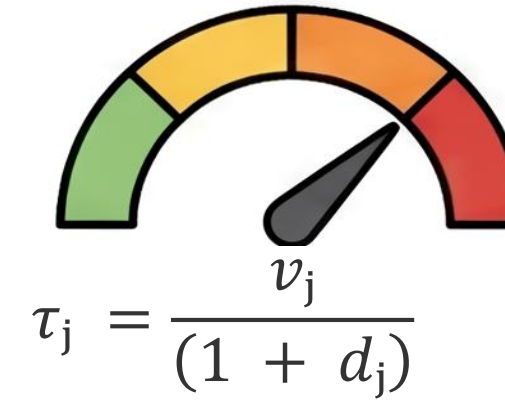
(b) Strategic: Value-Aware Commander (Every $T_{macro} = 10$ Steps)

Global Fire Asset Maps

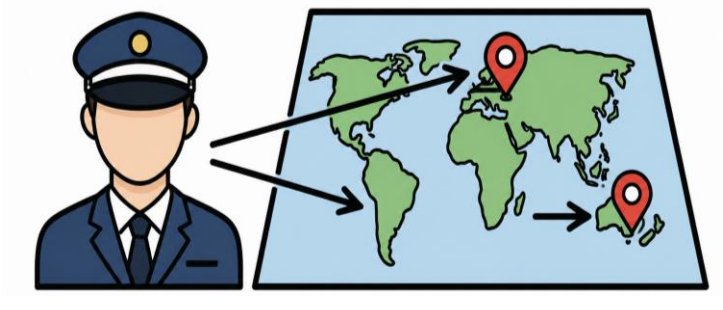


S_{fire}, S_{asset}

Threat Scoring



Assignment



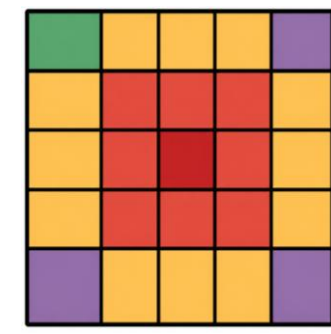
Rank by τ_j

Agents $\mapsto$ Top Ranked Threatened Assets

Assigned Target (y_i^*, x_i^*)

(c) Tactical: Learned Policy π_ϕ , Shared across agents, every step

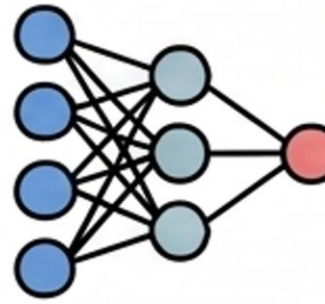
Egocentric Observation



$o_i^t \in \mathbb{R}^{9 \times 9 \times 8}$

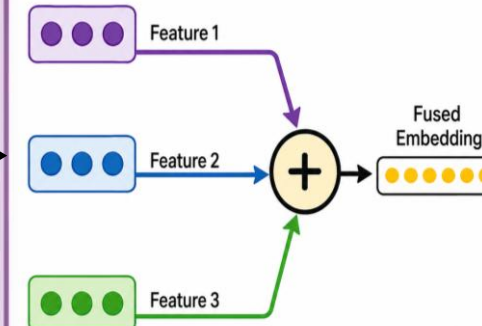
- Fire • Treated • Fuel
- Criticality Teammates
- Global Burn Fraction
- Target offset $(\Delta y_i, \Delta x_i)$

CNN Encoder



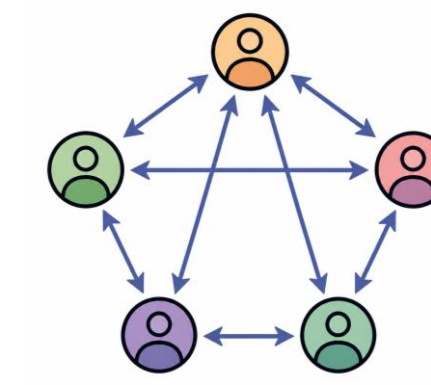
$h_i \in \mathbb{R}^{64}$

Target Fusion



$(\Delta y_i, \Delta x_i) \mapsto \mathbb{R}^{16}$

Communication



R=2 Round

$h_i \leftarrow \tanh(W_h h_i + W_c \bar{m}_{-i})$

Masked Action Head

a_i^t

Stay

$\uparrow \downarrow \leftarrow \rightarrow$

Treat

Joint Action $a_t = (a_t^1, \dots, a_t^{N_t})$

Simulator Advances 30 Simulated Minutes

Cross-Entropy on Expert Actions

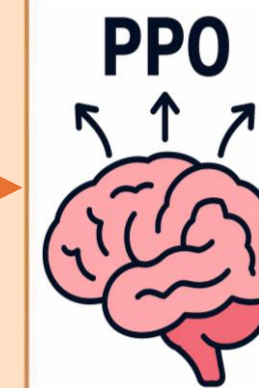
Clipped Policy Gradient

(d) Two-Phase Training (Commander Fixed)



Phase 1: BC warm-start

Clone value-aware asset-defense expert
(cross-entropy, 30 Episodes)



Phase 2: PPO fine-tuning

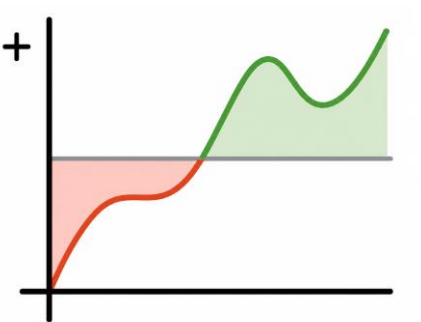
Centralized critic V_ψ

GAE

Clipping

Reward function

$r_t = -\Delta WEL + \alpha \Delta ISR + \eta \Delta burn$



Demonstrations (BC);
Rollouts and Reward r_t (PPO)